Associations Between Adverse Childhood Experiences and Adult Alcohol Use in the Behavioral Risk Factor Surveillance System (BRFSS) Dataset

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Abstract

The negative health effects of alcohol consumption, especially excessive consumption, make alcohol use and misuse a widespread public health concern. One proposed reason for excessive drinking is the self-medication hypothesis, which posits that people drink excessively to cope with negative emotions. In some cases, people may be motivated to self-medicate with alcohol due to the effects of adverse childhood experiences (ACEs) on mental health. The present study investigates the association between ACEs and alcohol use and misuse; specifically, it aims to identify whether people with ACEs drink more frequently and more heavily than people without ACEs. Two main statistical methods are used to evaluate these questions: a zero-inflated negative binomial model predicting alcohol use frequency and a binary logistic regression model predicting heavy drinking behavior. Results suggest that ACEs may be positively associated with alcohol misuse as operationalized by heavy drinking and that associations between ACEs and alcohol use frequency may vary by type of ACE, but these results are not statistically significant and therefore cannot be taken as conclusive empirical evidence. More research is needed to clarify and further elaborate on the presence, strength, and direction of findings.

Introduction

Literature Review

Alcohol consumption is a public health concern with major consequences for health and quality of life, with over 150,000 deaths per year associated with excessive drinking (CDC, 2025b). Although alcohol has long been assumed to be safe or even beneficial for health in small amounts, newer research is beginning to suggest that these amounts may be lower than we think or entirely nonexistent, potentially due to methodological flaws in previous research on the topic (Zhao et al., 2023). The CDC emphasizes that even moderate drinking confers some health risks and advises that people who do not currently drink refrain from starting (CDC, 2025a).

Accepted definitions of different levels of alcohol consumption according to the CDC (2025a, 2025b) are:

- Moderate Drinking = 2 drinks or fewer per day for men; 1 drink or fewer per day for women
- Binge Drinking: 5 drinks or more per occasion for men; 4 drinks or more per occasion for women
- Heavy Drinking: 15 drinks or more per week for men; 8 drinks or more per week for women

In the United States, over half of adults consume alcohol, 17% of adults engage in binge drinking, and 6% of adults engage in heavy drinking (CDC, 2025c). These data suggest that alcohol misuse is a major problem that contributes to poor health outcomes experienced by millions of Americans and emphasize the importance of understanding why people misuse alcohol.

One proposed reason for alcohol misuse that exists in both popular understanding and research literature is that some people drink to cope with negative feelings, an idea referred to as the self-medication hypothesis (Khantzian, 1997). One review of relevant literature found evidence in favor of the self-medication hypothesis specifically in the context of post-traumatic stress disorder (PTSD), although the authors note some methodological issues with studies included in the review that limit conclusions that can be drawn (Hawn et al., 2020). Another review found that self-medication rates ranged from approximately 22-24% in included studies, although the review did not differentiate between self-medication with alcohol and with other drugs (Turner et al., 2018). Although negative emotions are not the only cause of substance use disorders, existing literature suggests that their presence is associated with an increased risk for alcohol use and that it has the potential to turn into misuse or an alcohol use disorder if underlying issues are not managed (Turner et al., 2018).

Adverse childhood experiences (ACEs) are one way by which an individual can be exposed to an event that could cause negative emotions or psychological issues. ACEs can be loosely thought of as "potentially traumatic events that can have negative lasting effects on health and well-being" (Boullier & Blair, 2018, p. 132), specifically occurring in childhood. ACEs are associated with increased risk for a range of negative physical and mental health outcomes as well as for health risk behaviors (Boullier & Blair, 2018), of which alcohol misuse is one. Research has found that ACEs are associated with increased risk of alcohol misuse and that the likelihood of alcohol misuse increases as the number of distinct types of ACEs experienced increases (Anda et al., 2002; Dube et al., 2002; Fuller-Thomson et al., 2016; Loudermilk et al., 2018; Pilowsky et al., 2011; Sebalo et al., 2023). Given the negative effects that alcohol consumption, especially excessive alcohol consumption, has on health and the relative frequency at which ACEs occur, it is important that we understand the connection between ACEs and alcohol consumption in order to better understand and treat instances of alcohol misuse or alcohol use disorder.

Present Study

The research question that guides the present study is, "Are ACEs associated with alcohol consumption?" Specific aims are to investigate whether there is an association between (1) ACEs and any alcohol use, and (2) ACEs and alcohol misuse operationalized as heavy drinking. I expect to find similar results as previous research on this topic has established (e.g., Hughes et al., 2017); I anticipate that both alcohol misuse and alcohol use will be positively associated with ACEs. Specifically, I hypothesize that:

1. People with ACEs will drink alcohol more frequently than people without ACEs.
2. People with ACEs will be more likely to engage in heavy drinking than people without ACEs.
3. The predicted alcohol consumption frequency of people with ACEs will be higher than those of people without ACEs.
4. The predicted probability of engaging in heavy drinking for people with ACEs will be higher than those of people without ACEs.

The remainder of this paper focuses on the quantitative investigation of these hypotheses. First, I discuss the structure of the data and variables used in analyses. I then describe the analytic approaches taken, which include factor analysis and novel scale development, a zero-inflated negative binomial model that predicts alcohol use frequency, and a binary logistic regression model that predicts alcohol misuse as operationalized by heavy drinking. Finally, I present, interpret, and discuss the results of these three analytic approaches, as well as provide a summary of some methodological considerations and personal learning outcomes.

Method

Data

Data were sourced from the 2024 wave of the Behavioral Risk Factor Surveillance System (BRFSS; CDC, 2026). The BRFSS is a nationwide survey of adults that collects data on health risk behaviors, health conditions, and health-associated trends to support ongoing understanding of the state of health in the United States. As the survey collects data from all U.S. states and some U.S. territories, results from analyses conducted using the BRFSS data generalize to the entire adult U.S. population. Relevant to the present study are 13 questions that ask about respondents' exposure to different ACEs; full text of these survey questions can be found in Appendix A. 11 of these variables measure what I consider to be ACEs (e.g., "Did you live with anyone who was depressed, mentally ill, or suicidal?"), while the other two measure what I consider to be protective factors as they measure experiences that can be assumed to protect against development of negative outcomes (e.g., "For how much of your childhood was there an adult in your household who made you feel safe and protected?").

The full data of the 2024 BRFSS survey contains observations from $N = 457{,}670$ survey respondents. However, a large portion of respondents were either not asked the ACE-related questions or were marked as having missing data for these questions; these observations were dropped which resulted in a final sample size of $N = 38{,}748$. Demographic characteristics of the sample are presented in Table B1 in Appendix B.

Dependent Variables

Dependent variables used are alcohol use frequency (*DRNKANY6*; *M* = 4.2, *SD* = 7.7), operationalized as the number of days on which the respondent drank alcohol in the past 30 days, and heavy drinking (*_RFDRHV9*; *M* = 0.06, *SD* = 0.24), operationalized as whether or not the respondent engages in heavy drinking (defined as more than 14 drinks per week for males and more than 7 drinks per week for females). The alcohol use variable originally recorded a mix of counts of the number of days respondents drank alcohol in both the past week and the past month, so monthly alcohol use frequency estimates were calculated for respondents for whom weekly data was available by multiplying weekly drinks consumed by 4.

Alcohol use is a count variable, so I used a count model to model it; specifically a zero-inflated negative binomial model was selected because the Variance=Mean assumption does not hold for this variable (Variance-Mean Ratio = 14.2) and there is a large number of respondents that reported not drinking in the past month compared to respondents that reported all other counts of alcohol consumption days. Heavy drinking is a binary variable, and as such I used a binary logistic regression model to model it. There is no clustering evident in the data for either of these two variables that indicates the need for a multilevel model in either case. Distributions of the two dependent variables are displayed in Figure 1 (alcohol use) and Figure 2 (heavy drinking).

Figure 1

Distribution of Alcohol Use by BRFSS Respondents with Available ACE Question Data

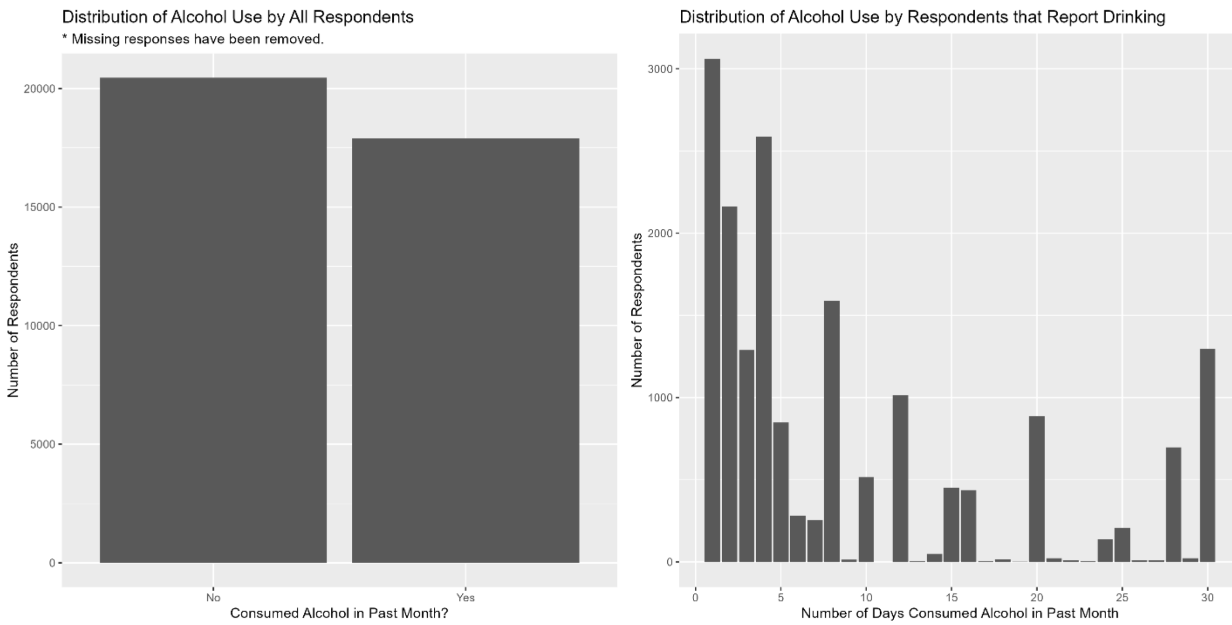
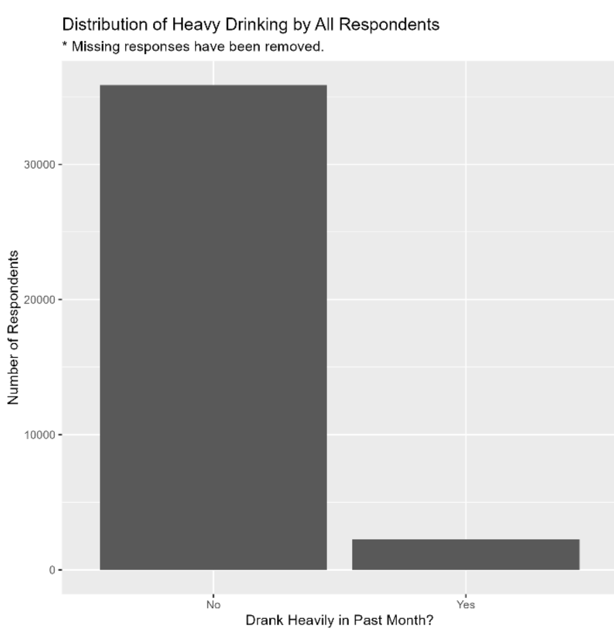


Figure 2

Distribution of Heavy Drinking by BRFSS Respondents with Available ACE Question Data



Independent Variables

Independent variables used are three novel ACE scales created from the 11 ACE variables using Principal Axis Factoring (PAF) with oblique rotation and validated using reliability testing (Cronbach's alpha). PAF was selected because the aim of the factor analysis was to create new composite variables, while oblique rotation was selected because I assume some correlation between factors due to their existing grouping into one overarching theoretical category (i.e., "ACE variables"). Detailed results of the factor analysis and reliability analysis are presented later in this paper in the Results section.

I chose to conduct factor analysis on the ACE variables and create new scales to reduce the number of primary independent variables in the models, which aims to aid with interpretability and minimize the risk of problems associated with using large numbers of predictors. Prior to any analysis, I converted each ACE variable to a binary scale where 0 indicates the respondent experienced the associated ACE and 1 indicates the respondent did not, to ensure that all ACE variables were measured on the same scale.

Control Variables

Control variables include a set of standard demographic variables (age, sex, sexual orientation, race, education level, income level, urban status, marital status, employment status, and veteran status), as well as two variables that measure physical and mental health status (*PHYSHLTH* and *MENTHLTH*) and the two aforementioned protective factor variables (*ACEADSAF* and *ACEADNED*). I included the health variables to control for differences between people in good health compared to people in poor health, as I believe there to be theoretical reasons that these groups may differ on alcohol use frequency and heavy drinking (e.g., people with poor mental health might drink more than people with better mental health due to self-medication attempts; people with poor physical health might drink less than people with good physical health because their health concerns prevent them from drinking). I include the two protective factor variables as control variables because I think they are theoretically different enough from the rest of the ACE variables to justify separation from them and because I believe they might impact development in a manner that warrants controlling for them (e.g., people who are exposed to protective factors may have better outcomes in general than people who are not and therefore may use or misuse alcohol less).

Results

Results of the factor analysis of the 11 ACE variables indicated that three distinct factors are present, which I name Sexual Abuse (*ACETOUCH*, *ACETTHEM*, *ACEHVSEX*), Other Abuse (*ACEPUNCH*, *ACEHURT1*, *ACESWEAR*), and Household Factors (*ACEDEPRS*, *ACEDRINK*, *ACEDRUGS*, *ACEPRISN*, *ACEDIVRC*). Fit statistics indicate that this factor analysis model fits the data well (Tucker-Lewis Index = 0.975; RMSEA = 0.035), while reliability calculations indicate that the Sexual Abuse scale ($\alpha = 0.81$, 95% CI [0.81-0.82]) has high reliability and the Other Abuse scale ($\alpha = 0.65$, 95% CI [0.64-0.66]) and the Household Factors scale ($\alpha = 0.63$, 95% CI [0.62-0.63]) have acceptable reliability with alpha scores only slightly below the rule-of-thumb threshold of $\alpha = 0.7$. The new scales were implemented by creating new binary variables that equal 0 if the respondent did not experience any of the ACEs captured by the scale and 1 if the respondent did experience any of the ACEs captured by the scale.

Model results (i.e., coefficients) for the three composite ACE scales are presented in Table 1; full model results for the remainder of the variables included in the models are presented in Table B2 in Appendix B. Prediction plots for the three composite ACE scales are presented in

Figure 3 (predicted number of days of alcohol use in the past month, by composite ACE scale) and Figure 4 (predicted probabilities of heavy drinking, by composite ACE scale).

Table 1

Results of Zero-Inflated Negative Binomial Model Predicting Alcohol Use and Binary Logistic Regression Model Predicting Heavy Drinking, Primary Independent Variables Only

	Dependent Variable:	
	Alcohol Use	Heavy Drinking
ACE: Sexual Abuse	0.062 (0.076)	0.041 (0.159)
ACE: Other Abuse	0.014 (0.048)	0.073 (0.107)
ACE: Household Factors	-0.088 (0.097)	0.187 (0.203)
Constant	1.492*** (0.196)	-3.855*** (0.437)
Observations	17,918	17,855
Log Likelihood	-39,859.180	-4,110.085
Akaike Inf. Crit.		8,328.170
Note:	*p<0.1; **p<0.05; ***p<0.01	

Figure 3

Prediction Plot for Alcohol Use by ACE Category

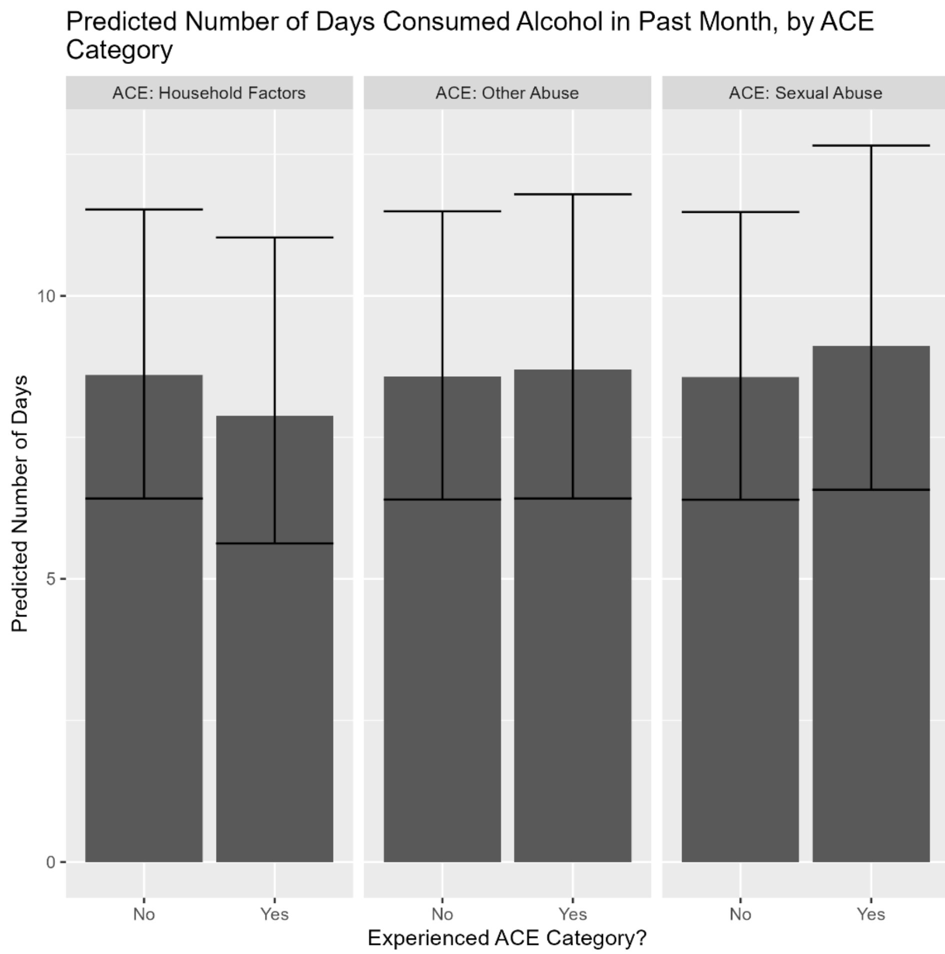
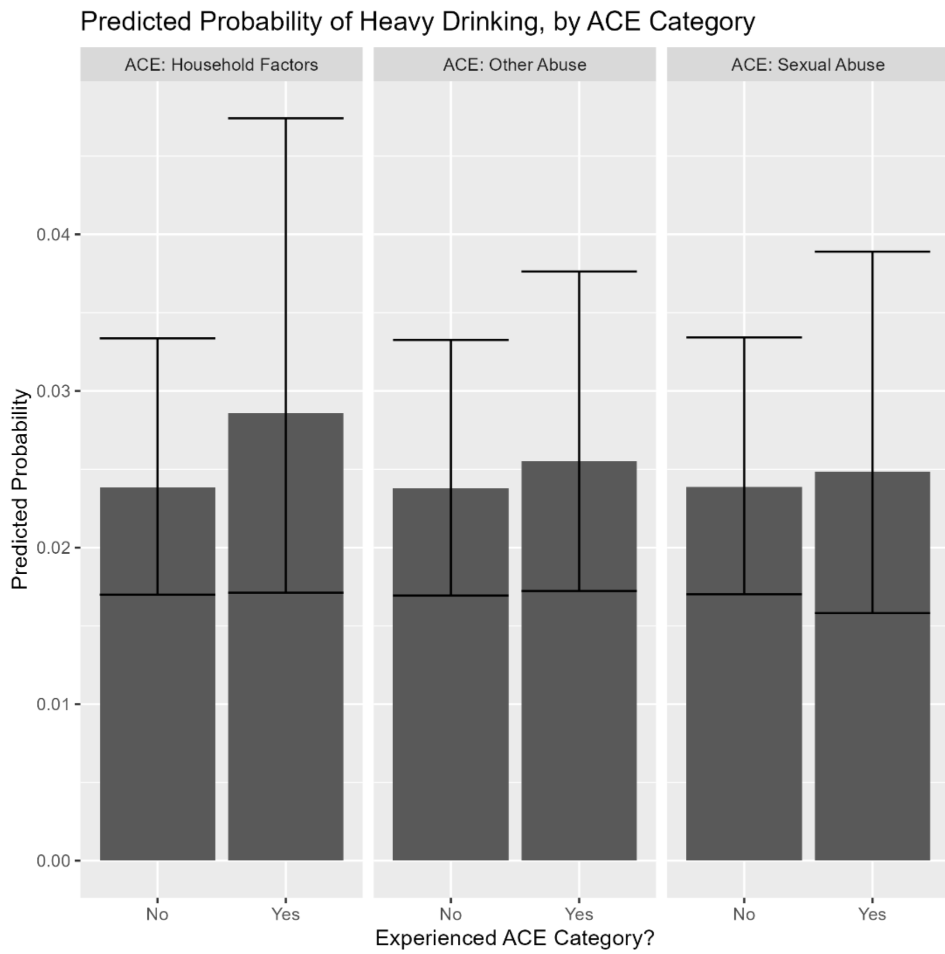


Figure 4

Prediction Plot for Heavy Drinking by ACE Category



To evaluate model fit, I compared each model to a simpler model within their own family on AIC. Comparison of the primary zero-inflated negative binomial model (AIC = 79936.36) to a comparison non-inflated negative binomial model (AIC = 81917.24) indicates that the primary count model for alcohol use performs better, and comparison of the primary binary logistic regression model with all predictors (AIC = 8328.17) to a comparison binary logistic regression model with no predictors (AIC = 17106.55) indicates that the primary logistic regression model for heavy drinking performs better. In other words, both of the primary models perform better than their simpler counterparts.

Discussion

Although none of my six main findings were statistically significant at the $p < 0.05$ level and therefore no statistical support for my hypotheses about alcohol use frequency and heavy drinking was found, I discuss my findings regardless to provide an overview of the patterns present in this data. It is also important to note that all six coefficients are relatively small in magnitude and have relatively wide confidence intervals, indicating that the direction and strength of any potentially present effect is additionally difficult to establish.

The count model for alcohol use frequency found that people with ACEs in the Household Factor category (i.e., parental mental illness or parental divorce) drank less frequently on average than people without ACEs ($\beta = -0.088$) and that people with ACEs in the Other Abuse category or the Sexual Abuse category drank more frequently than people without ACEs (Other Abuse $\beta = 0.014$; Sexual Abuse $\beta = 0.062$). Similarly, the logistic regression model for heavy drinking found that people who experienced ACEs in any of the three categories were, on average, more likely to drink heavily than people who did not experience that particular ACE type (Household Factors $\beta = 0.187$, Other Abuse $\beta = 0.073$; Sexual Abuse $\beta = 0.041$). However, to reiterate, none of these findings were statistically significant at $p < 0.05$, resulting in a low level of confidence in the veracity of these results.

Project Summary & Learning Reflection

The present study employs three different statistical approaches – factor analysis and new measure creation, a zero-inflated negative binomial model, and a binary logistic regression model – to look at associations between ACEs and later adult alcohol use. The zero-inflated negative binomial model was used to model alcohol use frequency because it is a count variable with a disproportionate number of zeros and the Variance=Mean assumption fails to be upheld, while the binary logistic regression model was used to model the heavy drinking variable because it is a dichotomous variable.

Results found no support for any of my hypotheses due to lack of statistical significance. However, the statistically non-significant patterns present in the data indicate that alcohol consumption frequency may be lower on average for people who experienced ACEs related to household factors and higher on average for people who experienced ACEs related to abuse, while probability of heavy drinking may be higher across the board with people who experienced each particular type of ACE being more likely to drink heavily than people who did not experience that particular type of ACE.

Through the process of completing this project, I developed a deeper understanding of the statistical methods I used and the manner in which research is communicated. I have used logistic regression for a similar research project at the intersection of psychology and public health in the past (predicting the presence of suicidality based on language use) but gained a better understanding of how the method works with this project. The previous project was closer to machine learning than statistics in nature and focused on maximizing prediction accuracy, while the present project was focused on interpretability of results and conclusions that could be drawn, which required a more detailed look at both methods and results. I also learned how best to communicate results from a project of this nature. I felt it best not to overwhelm readers with massive amounts of documentation and results in the main body of the paper and instead included only relevant items there, placing items like full regression tables and lists of survey questions in appendices.

I also learned about how alcohol use is associated with ACEs from both existing literature and my own analytic findings. While ideas like the self-medication hypothesis are compelling explanations for alcohol misuse, addiction, and alcohol-related health problems and deaths, the empirical support found for them in this project is shaky at best. The argument could be made that this might change with different and more rigorous methods than the observational and archival ones employed here, but it is still likely that a large number of phenomena influence something as broad as alcohol misuse and a research study of this nature will not be able to detect them all or untangle their effects.

Regardless, research of this sort still has a place in the broader context of science as knowledge generation. In the modern world, where data is abundantly available and communication of results is easier than ever before, analysts and researchers have the potential to cast wide nets into issues of public importance by conducting hobby projects or exploratory studies and communicating the results through unofficial channels. While a project like this lacks the institutional grounding of academic research, it is possible that discovering a project such as the present study could spark an idea in an individual who goes on to uncover important insights about health and behavior.

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Appendix A

Survey Questions Related to ACEs

Survey questions in the BRFSS that ask about respondents' history of experiences that are defined as ACEs, as well as their corresponding variable names as found in the raw dataset, are as follows:

1. *ACEDEPRS*: "Did you live with anyone who was depressed, mentally ill, or suicidal?"
2. *ACEDRINK*: "Did you live with anyone who was a problem drinker or alcoholic?"
3. *ACEDRUGS*: "Did you live with anyone who used illegal street drugs or who abused prescription medications?"
4. ACEPRISN: "Did you live with anyone who served time or was sentenced to serve time in a prison, jail, or other correctional facility?"
5. ACEDIVRC: "Were your parents separated or divorced?"
6. ACEPUNCH: "How often did your parents or adults in your home ever slap, hit, kick, punch or beat each other up?"
7. ACEHURT1: "Not including spanking, (before age 18), how often did a parent or adult in your home ever hit, beat, kick, or physically hurt you in any way?"
8. ACESWEAR: "How often did a parent or adult in your home ever swear at you, insult you, or put you down?"
9. ACETOUCH: "How often did anyone at least 5 years older than you or an adult, ever touch you sexually?"
10. ACETTHEM: "How often did anyone at least 5 years older than you or an adult, try to make you touch them sexually?"

11. ACEHVSEX: “How often did anyone at least 5 years older than you or an adult, force you to have sex?”
12. ACEADSAF: “For how much of your childhood was there an adult in your household who made you feel safe and protected?”
13. ACEADNED: “For how much of your childhood was there an adult in your household who tried hard to make sure your basic needs were met?”

Appendix B

Table B1

Demographic Characteristics of BRFSS Respondents with Available ACE Question Data

Variable:	*N* = 38,7481
Age (Years)	
18-24	1,971 (5.2%)
25-29	1,561 (4.1%)
30-34	1,896 (5.0%)
35-39	2,049 (5.4%)
40-44	2,219 (5.8%)
45-49	2,180 (5.7%)
50-54	2,577 (6.8%)
55-59	2,911 (7.6%)
60-64	3,813 (10%)
65-69	4,259 (11%)
70-74	4,403 (12%)
75-79	3,806 (10.0%)
80+	4,474 (12%)
Unknown	629
Sex	
Female	21,397 (55%)
Male	17,351 (45%)
Sexual Orientation	
Straight	22,498 (93%)
Gay	500 (2.1%)
Bisexual	709 (2.9%)
Other	411 (1.7%)
Unknown	14,630
Race	
White	22,793 (60%)
Black	4,142 (11%)
American Indian or Alaskan Native	332 (0.9%)
Asian	1,920 (5.0%)
Native Hawaiian or Pacific Islander	576 (1.5%)
Other Non-Hispanic	274 (0.7%)
Multiracial	1,583 (4.1%)
Hispanic	6,532 (17%)
Unknown	596

Education Level	
Did Not Graduate High School	2,316 (6.0%)
Graduated High School	9,732 (25%)
Some College or Technical School	10,773 (28%)
Graduated from College or Technical School	15,795 (41%)
Unknown	132
Income Level	
Less Than $15k	2,287 (7.2%)
$15k-$25k	3,273 (10%)
$25k-$35k	3,866 (12%)
$35k-$50k	4,396 (14%)
$50k-$100k	9,575 (30%)
$100k-$200k	6,404 (20%)
Greater Than $200k	2,091 (6.6%)
Unknown	6,856
Urban Status	
Rural	4,891 (15%)
Urban	28,549 (85%)
Unknown	5,308
Marital Status	
Married	18,738 (49%)
Divorced	5,275 (14%)
Widowed	5,109 (13%)
Separated	747 (1.9%)
Never Married	6,851 (18%)
Unmarried with Partner	1,763 (4.6%)
Unknown	265
Employment Status	
Employed	13,747 (36%)
Self-Employed	3,460 (9.0%)
Unemployed Greater Than 1 Year	761 (2.0%)
Unemployed Less Than 1 Year	734 (1.9%)
Homemaker	1,841 (4.8%)
Student	740 (1.9%)
Retired	14,785 (38%)
Unable to Work	2,382 (6.2%)
Unknown	298
Veteran Status	
Not a Veteran	33,302 (86%)
Veteran	5,368 (14%)
Unknown	78

Table B2

Results of Zero-Inflated Negative Binomial Model Predicting Alcohol Use and Binary Logistic Regression Model Predicting Heavy Drinking, for All Independent Variables

	Dependent Variable:	
	Alcohol Use	Heavy Drinking
ACE: Sexual Abuse	0.062 (0.076)	0.041 (0.159)
ACE: Other Abuse	0.014 (0.048)	0.073 (0.107)
ACE: Household Factors	-0.088 (0.097)	0.187 (0.203)
Protective Factor: Safe Adult	-0.139* (0.084)	0.181 (0.195)
Protective Factor: Needs Met	0.006 (0.136)	0.429 (0.336)
Physical Health	-0.003* (0.002)	-0.006 (0.004)
Mental Health	0.007*** (0.002)	0.017*** (0.004)
Age: 25-29	0.096 (0.086)	0.187 (0.213)
Age: 30-34	0.276*** (0.083)	0.485** (0.198)
Age: 35-39	0.316*** (0.085)	0.533*** (0.200)
Age: 40-44	0.385*** (0.085)	0.548*** (0.198)
Age: 45-49	0.412*** (0.087)	0.421** (0.206)
Age: 50-54	0.405*** (0.087)	0.502** (0.203)
Age: 55-59	0.433*** (0.086)	0.300 (0.207)
Age: 60-64	0.569*** (0.084)	0.580*** (0.198)
Age: 65-69	0.643*** (0.087)	0.320 (0.208)
Age: 70-74	0.618*** (0.089)	0.120 (0.217)
Age: 75-79	0.768*** (0.094)	0.129 (0.226)
Age: 80+	0.806*** (0.099)	-0.585** (0.255)
Sex: Male	0.358*** (0.029)	0.159** (0.068)
Sexual Orientation: Gay	-0.140 (0.087)	-0.432* (0.231)
Sexual Orientation: Bisexual	0.163** (0.073)	0.381** (0.155)
Sexual Orientation: Other	0.264** (0.114)	0.304 (0.227)
Race: Black	-0.272*** (0.048)	-0.714*** (0.130)
Race: American Indian or Alaskan Native	-0.057 (0.143)	0.238 (0.256)
Race: Asian	-0.287*** (0.058)	-0.594*** (0.147)
Race: Native Hawaiian or Pacific Islander	-0.027 (0.096)	0.320* (0.167)
Race: Other Non-Hispanic	-0.035 (0.170)	-0.045 (0.396)
Race: Multiracial	-0.052 (0.058)	0.049 (0.120)
Race: Hispanic	-0.207*** (0.059)	-0.412*** (0.147)
Education: Graduated High School	-0.244*** (0.092)	-0.116 (0.171)
Education: Some College or Technical School	-0.312*** (0.092)	-0.249 (0.173)
Education: Graduated from College or Technical School	-0.273*** (0.092)	-0.399** (0.174)
Income: 15k-25k	-0.152 (0.106)	-0.270 (0.208)
Income: 25k-35k	-0.057 (0.103)	0.083 (0.192)
Income: 35k-50k	-0.055 (0.100)	0.008 (0.191)

Income: 50k-100k	-0.025 (0.096)	0.117 (0.181)
Income: 100k-200k	0.019 (0.098)	0.253 (0.190)
Income: Greater Than 200k	0.144 (0.104)	0.596^{***} (0.206)
Urban/Rural Status: Urban	0.165^{***} (0.042)	0.203^{**} (0.100)
Marital Status: Divorced	0.110^{**} (0.043)	0.374^{***} (0.094)
Marital Status: Widowed	-0.001 (0.055)	-0.018 (0.134)
Marital Status: Separated	0.187^{*} (0.110)	0.302 (0.228)
Marital Status: Never Married	0.077^{*} (0.044)	0.326^{***} (0.098)
Marital Status: Unmarried with partner	0.118^{*} (0.066)	0.363^{**} (0.150)
Employment Status: Self-Employed	0.164^{***} (0.045)	0.222^{**} (0.102)
Employment Status: Unemployed Greater Than 1 Year	0.025 (0.120)	-0.258 (0.257)
Employment Status: Unemployed Less Than 1 Year	0.182^{*} (0.099)	0.562^{***} (0.187)
Employment Status: Homemaker	0.064 (0.086)	-0.138 (0.203)
Employment Status: Student	-0.170 (0.114)	-0.324 (0.287)
Employment Status: Retired	0.103^{**} (0.046)	0.151 (0.110)
Employment Status: Unable to Work	0.002 (0.093)	-0.525^{***} (0.196)
Veteran Status: Veteran	-0.043 (0.039)	-0.044 (0.095)
Constant	1.492^{***} (0.196)	-3.855^{***} (0.437)
Observations	17,918	17,855
Log Likelihood	-39,859.180	-4,110.085
Akaike Inf. Crit.		8,328.170
Note:	$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$	